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March 5, 2025

To:
Mr. Michael J.K. Kratsios
Assistant to the President for Science and Technology
Director, Office of Science and Technology Policy

Dear Mr. Kratsios,

In light of the recent release of a near-SotA AI model (R1) from the Chinese company DeepSeek, I am writing to propose some concrete actions the Administration can consider taking in response to these developments, as part of the nation's AI Action Plan.

As you are likely aware, DeepSeek's models stand out for several key reasons:

- They are all released under the MIT License with “open weights” and most with accompanying technical reports detailing their implementation freely available on arXiv. This lowers the bar for setting up local deployment of the models as well as their unrestricted modification.
- R1's performance on a variety of benchmarks is nearly on par with SoTA LLMs like OpenAI's GPT-4o and o1, despite being a significantly smaller model by number of parameters and cost of inference.
- DeepSeek is a small startup with under 200 employees, only founded in July 2023. According to the company, its V3 model cost only \$6 million to train, compared to \$100 million for OpenAI's GPT-4, and used about one-tenth of the compute used to train Meta's comparable Llama 3.1 model.
- DeepSeek achieved this in spite of the US's export controls on advanced Nvidia GPUs, forcing them to use an older, less powerful chipset.

All of these factors contributed to the massive \$1 trillion loss in the market value of US tech companies, with Nvidia alone accounting for \$600 billion of that. It is clearly a moment to pause and reevaluate our national AI research and development strategy.

Days before the release of R1, the President announced the Stargate Project, a \$500 billion AI infrastructure investment by and into the private tech sector, led by OpenAI, Oracle, and SoftBank. Much of that is planned to go into building more and larger data centers, the idea or hope being that scaling laws will continue to hold.

But DeepSeek calls those assumptions into question. As figures like Gary Marcus have vocally expressed for years now¹, outside of Silicon Valley many AI experts and academics are skeptical

¹ <https://open.substack.com/pub/garymarcus/p/breaking-news-scaling-will-never>

that the scaling paradigm will be what leads to the next breakthrough in AI, let alone AGI. The compute-quality correlation is an empirically observed phenomenon, with no theoretical basis as of yet. Therefore, there's no guarantee that higher compute won't hit some plateau in quality (as OpenAI's new GPT-4.5 is beginning to show), nor that we can't achieve the same quality with less compute, as DeepSeek suggests (as do techniques like distillation).

Thus, my proposal is that we begin considering realigning both funding and research efforts somewhat in the next few months. The National Artificial Intelligence Research Resource (NAIRR) Pilot launched last year is one such avenue. For a fraction of the investment needed to build massive new data centers, I propose allocating a portion of the NSF appropriations into establishing a new graduate fellowship program focused on AI, similar to the DoE NDSEG or CSGF, and complementing the existing NSF GRFP (of which I was a fellow). In particular, there should be equal emphasis on theory and foundational research as on the business and industrial applications of AI.

As the first holder of your office, Vannevar Bush, wrote in his epochal 1945 report²:
“Progress ... depends upon a flow of new scientific knowledge. ... This essential, new knowledge can be obtained only through basic scientific research. ... With some notable exceptions, most research in industry and Government involves application of existing scientific knowledge to practical problems. It is only the colleges, universities, and a few research institutes that devote most of their research efforts to expanding the frontiers of knowledge.”

It would serve us well to remember and heed Bush's advice. No doubt there will be pushback from industry, as increased investment into basic science will in the short term likely compete with funding for their own product-driven research efforts. But in the long run, as we have seen time and time again, from the Internet to mRNA vaccines, the return on investment of basic research is nigh incalculable.

Very respectfully submitted,

Haihao Liu
Founder, Lead AI



² https://nsf-gov-resources.nsf.gov/2023-04/EndlessFrontier75th_w.pdf